Gavin's Lectures on Physics - Condensed Matter



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July 24, 2026

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1 Foundations of Phase Transitions and Statistical Mechanics

1.1 Phase Transition

1.1.1 Discontinuous or First-order Phase Transition

Two distinct phases coexist at phase transition. Latent (潜在的) Heat associated with transition.

1.1.2 Continuous or second-Order Phase Transition

Two phases are indistinguishable at transition. No latent heat, but correlation length diverges at transition. system components are correlated in a universal way at transition criticality (临界点).

1.2 Ensemble

In canonical ensemble: N, V, T, fixed. Potential : $F = E - TS \quad (E = U)$ Minimized F:

- High T : S dominates, disordered system.
- Low T : E dominates, ordered system.

1.3 Thermodynamic Limit

$$N \to \infty, \quad V \to \infty, \quad n = \frac{N}{V} = \text{constant} \tag{1}$$

1.4 Symmetry

1.4.1 Basic symmetries

- Translation symmetry : momentum conservation
- Rotation symmetry : angular momentum conservation.
- Inversion symmetry
- Time-reversal symmetry
- Gauge symmetry.

1.4.2 Spontaneous (自发的) Symmetry Breaking (SSB)

Def: The Hamiltonian of a system processes a certain symmetry, but the equilibrium ground state spontaneously loses part of this symmetry.

1.4.3 Order Parameter

Def: A measurable physical quantity that quantifies the degree of symmetry breaking.

$$\text{Order Parameter} \begin{cases} = 0 & , \text{ high symmetry, disordered phases} \\ \neq 0 & , \text{ low symmetry, ordered phases.} \end{cases} \tag{2}$$

2 Disorder, Long-Range Order, and Universality

2.1 Disorder

2.1.1 Domain (畴)

1. Impurities (杂质) introduce local symmetry breaking fields, leading to magnetic/structural domains.

2. The formation of domains depends on the competition between domain wall energy cost and energy benefit from impurities.

3. Energy Scaling . (L: system size, d: dimension)

2.1.2 Existence of Long-Range Order (LRO) in the Thermodynamic Limit

Dimension	Discrete (e.g., Ising)	Continuous (e.g., XY)
1D	No LRO ($T > 0$)	No LRO (Mermin-Wagner)
2D	LRO exists ($T < T_c$, Onsager solution)	No LRO, but quasi-LRO ($T < T_{\mathrm{BKT}}$)
3D	LRO exists	LRO exists ($T < T_c$)

LRO: (Long-Range order). Correlations between microscopic degrees of freedom persist at macroscopic / infinitely large distance.

2.1.3 Mermin–Wagner Theorem

For systems with **continuous symmetry and short-range interactions**, no spontaneous symmetry breaking, and no long-range order exist at finite temperature in 1D and 2D systems. **Notably, the 2D XY model allows for quasi-long-range order (BKT phase), which is an exception to conventional LRO.**

2.2 Universality

2.2.1 Universality Class

A universality class is a group of distinct physical systems that share the same set of key global properties: identical critical exponents, scaling functions, critical behaviors at critical points.

2.2.2 Universality Principle

Near Critical Point T_C, microscopic details become irrelevant. Different systems exhibit identical critical behaviours and belong to the same universality class.

A universality class is determined only by system dimension, order parameter dimension, and symmetry.

2.2.3 Critical Exponents

Power-law behaviour near critical point T_C.

1. order parameter : $\Delta \sim |T - T_C|^{\beta}$
2. magnetic susceptibility (磁化率) : $\chi \sim |T - T_C|^{-\gamma}$
3. Correlation length : $\xi \sim |T - T_C|^{-\nu}$

3 Renormalization Group (RG) (重整化群)

3.1 Parameters

- Relevant parameters : Grow under RG transformation, dominate macroscopic long-wavelength behaviours.
- Irrelevant parameters : Decay under RG transformation, negligible for macroscopic properties.

3.2 Standard RG Procedures

1. Coarse-graining (粗粒化): Combine adjacent microscopic degrees of freedom into blocks.
2. Rescaling (尺度重标): shrink all spatial lengths by a scaling factor $b > 1$.
3. Renormalization (参数重整): Rescale physical parameters and order parameters to restore the original system form.

3.3 RG fixed point

Parameters keep invariant under RG transformation

- stable fixed point : corresponds to stable thermodynamic phase .
- Unstable fixed point : Corresponds to phase transition critical point.

3.4 RG flow

Evolution & trajectory of parameters during scale transformation .

4 Landau Theory of Phase Transitions

4.1 Theory

1. Phenomenological theory constrained by system symmetry.
2. Ignores thermal fluctuations. (MFT 平均场理论)
3. Expand free energy F, as power series of order parameter Δ.

$$\Delta \to \begin{cases} S \\ E \end{cases} \Rightarrow F. \tag{3}$$

4.2 Second-Order Phase Transition

4.2.1 Free energy formula

$$F(\Delta) = F_0 + r_0 \left(T - T_C\right) \Delta^2 + s_0 \Delta^4 \quad (r_0, s_0 > 0) \tag{4}$$

No odd-term due to symmetry constraint.

$$\begin{cases} T > T_C;\ F \text{ minimum at } \Delta = 0, \text{ disordered phase} \\ T < T_C:\ F \text{ minimum at } \Delta = \pm\sqrt{\frac{r_0(T_C - T)}{2s_0}}, \text{ ordered phase.} \end{cases} \tag{5}$$

4.2.2 Mean-field critical exponents

$$\begin{cases} \Delta \sim |T - T_C|^{\beta} \quad , \ \beta = \frac{1}{2} \\ \chi \sim |T - T_C|^{-\gamma} \quad , \ \gamma = 1 \\ \xi \sim |T - T_C|^{-\nu} \quad , \ \nu = \frac{1}{2} \end{cases} \tag{6}$$

4.3 First-order Phase Transition

4.3.1 Free energy formula

$$F(\Delta) = F_0 + a(T - T_0)\Delta^2 - b\Delta^4 + c\Delta^6 \quad (a, b, c > 0) \tag{7}$$

Order parameter jumps discontinuously.

4.3.2 Features

Latent heat, supercooling & superheating, discontinuous order parameter.

4.4 Ginzburg Criterion

MFT: mean field theory . (Landau theory $\in$ MFT)

$$G = \frac{\text{thermal fluctuation strength}}{\text{ordering strength}} \tag{8}$$

$$\begin{cases} G \ll 1 \quad : \text{ MFT valid} \\ G \gtrsim 1 \quad : \text{ MFT fails} \end{cases} \tag{9}$$

Dimension :

$$\begin{cases} d > 4 & ; \text{ MFT holds for all } T \\ d = 4 & ; \text{ Marginal} \\ d < 4 & ; \text{ MFT works far from } T_C, \text{ breaks down in critical region, requiring RG treatment.} \end{cases} \tag{10}$$

5 Ising Model

Discrete symmetry

5.1 Ising Hamiltonian

$$H = -J\sum_{\langle ij\rangle} s_i s_j - h\sum_i s_i \tag{11}$$

$$\begin{cases} s_i = \pm 1, \quad \text{Lattice (格点) spin} \\ J : \text{Exchange coupling.} \begin{cases} J > 0 & : \text{ ferromagnetism} \\ J < 0 & : \text{ anti-ferromagnetism} \end{cases} \\ h : \text{external magnetic field, } h = \mu B \\ \langle ij\rangle : \text{nearest-neighbour pairs} \end{cases} \tag{12}$$

5.2 Self-Consistent Equation

5.2.1 Mean-Field Decoupling

In mean-field approximation:

$$\sum_{\langle ij\rangle} s_i s_j \approx \sum_{\langle ij\rangle}[m(s_i+s_j)-m^2] = \sum_i qms_i - \frac{1}{2}Nqm^2 = Nqm^2 - \frac{1}{2}Nqm^2 = \frac{1}{2}Nqm^2 \tag{13}$$

$$H = -J\sum_{\langle ij\rangle} s_i s_j - h\sum_i s_i \tag{14}$$

$$= -\frac{1}{2}\sum_{ij} J_{ij}\, s_i s_j - h\sum_i s_i, \quad J_{ij} = \begin{cases} J & ,\langle i,j\rangle \text{ near} \\ 0 & ,\langle i,j\rangle \text{ far.} \end{cases} \tag{15}$$

$$\Rightarrow \frac{\partial H}{\partial s_i} = -\sum_{j,\text{near } i} J_{ij} s_j - h = -Jq\langle s\rangle - h \tag{16}$$

Def: $\bar{h} = -\frac{\partial H}{\partial s_i} = Jq\langle s\rangle + h \Rightarrow H = \frac{1}{2}NJqm^2 - \bar{h}\sum_i s_i$

5.2.2 Partition Function and Magnetization

$$Z = \sum_{s_1=\pm1}\sum_{s_2=\pm1}\cdots\sum_{s_N=\pm1} e^{-\beta H} \tag{17}$$

$$H \approx -\big(Jq\langle s\rangle + h\big)\sum_i s_i = -\bar{h}\sum_i s_i \tag{18}$$

$$\Rightarrow \begin{cases} Z = \prod_{i=1}^N \sum_{s_i=\pm1} e^{\beta\bar{h}s_i} = \prod_{i=1}^N Z_1 \\ Z_1 = e^{\beta\bar{h}} + e^{-\beta\bar{h}} \end{cases} \tag{19}$$

5.2.3 Self-Consistent Equation

$dQ = 0$

$$\Rightarrow dW = dU = d\left(-\bar{h}\sum_i s_i\right) = d\left(-N\bar{h}\langle s\rangle\right) = -N\langle s\rangle d\bar{h} \tag{20}$$

$$\because\ dW = Ydy, \quad Y = -\frac{N}{\beta}\frac{\partial \ln Z_1}{\partial y} \tag{21}$$

$$\therefore\ N\langle s\rangle = \frac{N}{\beta}\frac{\partial \ln Z_1}{\partial y} \tag{22}$$

$$\therefore \langle s\rangle = \frac{1}{\beta}\frac{\partial \ln Z_1}{\partial y} = \tanh(\beta\bar{h}) \tag{23}$$

Here,$\beta = \frac{1}{k_B T}$

$$m = \langle s\rangle \Rightarrow m = \tanh\left(\frac{Jqm + h}{k_B T}\right) \tag{24}$$

5.2.4 Critical Temperature

Without external field: $h = 0$:

$$\begin{cases} m = \tanh\left(\frac{Jqm}{k_B T}\right) \\ \text{Def}:\ T_C = \frac{Jq}{k_B} \end{cases} \Rightarrow m = \tanh\left(\frac{T_C}{T}\cdot m\right) \tag{25}$$

$$\begin{cases} T \ll T_C & : m = \tanh\left(\frac{T_C}{T}m\right) \to 1,\ \text{ordered system} \\ T \gg T_C & : m = \tanh\left(\frac{T_C}{T}m\right) \approx \frac{T_C}{T}m \Rightarrow m = 0,\ \text{disordered system} \end{cases} \tag{26}$$

5.2.5 Compare Mean-Field Theory with Exact Results

Dimension	Mean-Field Theory (MFT) Prediction	Exact Result
1D	$T_C = 2J/k_B$	$T_C = 0$
2D	$T_C = 4J/k_B$	$T_C \approx 2.269J/k_B$ (Onsager)
3D	$T_C = 6J/k_B$	$T_C \approx 4.511J/k_B$

5.3 Bragg-Williams Free Energy

5.3.1 Entropy from Combinatorics

$$\begin{cases} m = \frac{N_+ - N_-}{N} \\ N = N_+ + N_- \end{cases} \Rightarrow \begin{cases} N_+ = \frac{(1+m)N}{2} \\ N_- = \frac{(1-m)N}{2} \end{cases} \tag{27}$$

Stirling : $\ln(m!) \approx m\ln m - m$.

$$\therefore\ \Omega = C_N^{N_+} = \frac{N!}{N_+!\,N_-!} \tag{28}$$

$$\begin{aligned}\ln \Omega &= \ln \frac{N!}{N_+! \, N_-!} \\ &\approx N \ln 2 - \frac{N(1+m)}{2} \ln(1+m) - \frac{N(1-m)}{2} \ln(1-m)\end{aligned} \tag{29}$$

$$S = k_B \ln \Omega \tag{30}$$

5.3.2 Internal Energy

$$U = H = \frac{1}{2} N J q m^2 - \bar{h} \sum_i s_i = \frac{1}{2} N J q m^2 - \bar{h} N m \tag{31}$$

$$(\bar{h} = Jqm + h) \tag{32}$$

$$\Rightarrow U = -\frac{1}{2} N J q m^2 - N h m \tag{33}$$

5.3.3 Free Energy Formula

$$F = U - TS \tag{34}$$

$$= -\frac{1}{2} N J q m^2 - N h m - k_B T \ln \Omega \tag{35}$$

$$= N \left\{ -\frac{1}{2} J q m^2 - h m - k_B T \left[\ln 2 - \frac{1+m}{2} \ln(1+m) - \frac{1-m}{2} \ln(1-m) \right] \right\} \tag{36}$$

5.4 Verify Self-Consistent Equation

To minimize free energy:

$$\frac{dF}{dm} = 0 \tag{37}$$

We can derive Self-Consistent Equation:

$$\therefore \; m = \tanh \left(\frac{Jqm + h}{k_B T} \right) \text{ minimizes free energy } F(m) \tag{38}$$

6 XY Model

Continuous symmetry, 2D

6.1 XY Hamiltonian

$$H = -J \sum_{\langle i,j \rangle} \cos\left(\theta_i - \theta_j\right) \tag{39}$$

$$\begin{cases} \theta_i: \text{ spin azimuthal (方位的) angle} \\ J: \text{ exchange coupling} \\ \langle i,j \rangle: \text{ nearest-neighbour pairs} \end{cases} \tag{40}$$

Long range fluctuation : $|\theta_i - \theta_j| \ll 1$

$$\cos(\theta_i - \theta_j) \approx 1 - \frac{1}{2}\left(\theta_i - \theta_j\right)^2 \tag{41}$$

$$H \approx H_0 + \frac{1}{2}\left(\theta_i - \theta_j\right)^2 \tag{42}$$

$$\approx H_0 + \frac{J}{2}\int d^2x \left|\nabla\theta(\mathbf{x})\right|^2 = H_0 + H_V \tag{43}$$

6.2 Topological Defects (拓扑缺陷)

Topological Charge :

$$n = \frac{1}{2\pi}\oint_C \nabla\theta \cdot d\mathbf{r} \quad \in \mathbb{Z} \tag{44}$$

C: closed contour enclosing the defect core.

- Vortex (涡旋): spins rotate counterclockwise by 2π when circling the defect core , $n = +1$
- Anti-vortex (反涡旋): spins rotate clockwise by 2π when circling the defect core , $n = -1$.

6.3 Energy of single vortex

single vortex:

$$1 = n = \frac{1}{2\pi}\oint_C \nabla\theta \cdot d\mathbf{r} = |\nabla\theta| \cdot \frac{1}{2\pi}\oint_C dr = |\nabla\theta| \cdot \frac{1}{2\pi} \cdot 2\pi r \tag{45}$$

$$\Rightarrow |\nabla\theta| = \frac{1}{r} \tag{46}$$

$$\Rightarrow \int d^2x \left|\nabla\theta\right|^2 = \int_a^L \frac{1}{r^2} \cdot 2\pi r \, dr = 2\pi \ln\left(\frac{L}{a}\right) \tag{47}$$

$$\Rightarrow E_V = H_V = \frac{J}{2}\int d^2x \left|\nabla\theta\right|^2 = \pi J \ln\left(\frac{L}{a}\right) \tag{48}$$

Large system : $L \gg a$: $E_V \approx \pi J \ln L$ L: system radius. a: cell core radius

6.4 Vortex Entropy

$$\Omega \propto \frac{L^2}{a^2} \tag{49}$$

$$S = k_B \ln \Omega = 2k_B \ln \left(\frac{L}{a} \right) \tag{50}$$

Large system ($L \gg a$):

$$S \approx 2k_B \ln L \tag{51}$$

6.5 BKT Transition

6.5.1 Definition

BKT Transition : topological phase transition driven by the binding and unbinding of vortex–antivortex pairs. It occurs without symmetry breaking .

6.5.2 Free Energy ($L \gg a$)

$$\begin{cases} F_V = E_V - TS_V \\ E_V \approx \pi J \ln L \\ S_V \approx 2k_B \ln L \end{cases} \Rightarrow F_V = (\pi J - 2k_B T) \ln L \tag{52}$$

$$\begin{cases} F_V = (T_C - T)\, 2k_B \ln L \\ T_C = \frac{\pi J}{2k_B} \quad \text{critical temperature.} \end{cases} \tag{53}$$

$$\begin{cases} T < T_C \quad : \ F_V > 0, \ \text{Isolated vortices are unstable} \\ \qquad\quad \text{Vortex–antivortex pairs are tightly bound.} \\ T > T_C \quad : \ F_V < 0, \ \text{Isolated vortices are stable} \\ \qquad\quad \text{Vortex–antivortex pairs dissociate into free vortices.} \end{cases} \tag{54}$$

6.5.3 Correlation Function Behavior

- $T < T_C$: Quasi-long-range order (准长程序) exists.
- $T > T_C$: Only short-range order exists.

7 Hopfield Neural Network

7.1 Definition

A recurrent artificial neural network based on the energy function of Ising model.

7.2 Network Energy Function

$$H = -\frac{1}{2}\sum_{i \neq j} J_{ij}\, s_i s_j \tag{55}$$

$$\begin{cases} s_i = \pm 1 : \text{Neuron state. } (+1 : \text{Active, } -1 : \text{inactive}) \\ J_{ij} : \text{synaptic weight (突触权重)} \end{cases} \tag{56}$$

7.3 Hebbian Learning Rule

$J_{ii} = 0$ for all i.
Symmetric weight matrix :

$$J_{ij} = \frac{1}{N}\sum_{\mu=1}^{P} \xi_i^\mu \xi_j^\mu \tag{57}$$

$$\begin{cases} \xi_i^\mu : \text{ stored pattern, } i\text{-th component of the } \mu\text{-th pattern.} \\ \xi_i^\mu = \pm 1 \\ N : \text{ total number of neurons (神经元)} \\ P : \text{ number of stored patterns} \end{cases} \tag{58}$$

7.4 Neuron Update Rule

$$s_i(t+1) = \text{sign}\left(\sum_j J_{ij}\, s_j(t)\right) \tag{59}$$